Mallurwar Varad Nandkishor

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.27
Disha Delphi Public School, Kota	Class XII (CBSE)	2023	91.00%
Dnyanprakash Vidyaniketan, Latur	Class X (Maharshtra State Board)	2021	99.60%

SCHOLASTIC ACHIEVEMENTS

- **NTSE Scholar** - Secured Top 5 rank in Maharashtra State Level Examination and cleared the National Level Examination.
- **Gold Medalist** -Dr. Homi Bhabha Young Scientist Competition
- **State Rank 3** - Maharashtra State Scholarship Exam(Class 8)
- **International Rank 28** - SOF IMO (Class 7)
- **State Rank 5** - Maharashtra State Scholarship Exam(Class 4)

SKILLS

- **Programming Languages** : C, C++, Python.
- **Technical Skills** : DSA,Object Oriented Programming, ROS2(C++ packages, workspace setup, launch file creation, basic use of colcon build, and CMakeLists.txt configuration), Machine Learning, Problem solving.
- **Tools and Frameworks** : Matlab, Gazebo, Rviz, Latex, Solid Edge, Fusion 360.
- **Soft Skills** : Adaptability, Time Management, Networking, Teamwork, Leadership, Communication.

PROJECTS

- **Smart Room Automation: Object-Oriented Device Management System**
- **Tech Stack: C++**, **OOP**, STL (vector, unique_ptr), CMake
- Built a menu-driven multi-room simulator for device management, individual controls, and room-wide automation.
- Applied **abstract classes** and **polymorphism** through Device, Fan, and Light implementations.
- Used std::unique_ptr for safe SmartHome -> Room -> Device ownership and STL for state management.
- Implemented validated fan-speed and light-brightness controls, unique names, and status/error reporting.
- Separated console UI from application logic using SmartHomeController.
- **Concepts: abstraction, encapsulation, inheritance, runtime polymorphism**, **composition, RAII, smart pointers, STL, validation, separation of concerns.**

- **RAG AI Agent: PDF & Web Knowledge Retrieval**
- **Tech Stack: Python, LangChain, Pinecone, Hugging Face, Tavily API, FastAPI, Streamlit**
- Built an end-to-end RAG system querying both ingested PDFs and live web data.
- Embedded document chunks via Hugging Face and indexed vectors into Pinecone for low-latency retrieval.
- Integrated Tavily API to ground responses in real-time web search and minimize hallucinations.
- Developed a FastAPI backend paired with an interactive Streamlit UI for seamless query handling.
- **Concepts: RAG, vector search, semantic embeddings.**

- **Image Caption Generator: CNN-LSTM Pipeline**
- **Tech Stack: Python, TensorFlow, Keras, VGG16, LSTM, NLTK**
- Built an end-to-end deep learning model to generate descriptive captions for input images.
- Extracted high-level image features using a pre-trained VGG16 network.
- Implemented a dual-branch neural network merging visual features with an LSTM text decoder.
- Designed a custom batch data generator to train progressive text sequences without RAM overflow.
- Assessed caption quality across validation images using standard BLEU score benchmarks.
- **Concepts: Computer vision, sequence modeling, data generators, transfer learning, model evaluation.**

EXPERIENCE

- **Technology Summer Intern, CHUBB : Interview Intelligence Engine**
- **Tech Stack: Python, FastAPI, LLMs, MongoDB, Angular**
- Developed a modular real-time interview engine that dynamically adapts questioning based on candidate responses and evolving interview state.
- Designed an **Answer Analyzer -> State Updater -> Decision Engine -> Question Generator** pipeline with Interview State as the single source of truth.
- Built structured state tracking for skill progress, evidence, objectives, and interview context while separating objective selection from natural-language question generation.
- **Concepts: LLM orchestration, adaptive interviewing, state-driven architecture, prompt engineering, modular backend design.**

RELEVANT COURSES

- **Computer Science** : Introduction to Programming, Computer Architecture, Database Management Systems

- **Math** : Elementary Linear Algebra, Differential Equations, Transform Techniques, Probability, Complex Variables, Statistics, Calulus-1, Calculus-2

- **Miscellaneous** : Introduction to Modern AI, Introduction to Entrepreneurship, Communication Skills